

CONCEPTS OF FORCE

Definition: A force is an interaction that causes an object to change its motion, speed, or direction. It can be described as a push or a pull on an object.

Units of Measurement

The standard unit of force is the Newton (N), defined as the force required to accelerate a one-kilogram mass by one meter per second squared ($1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2$).

Types of Forces

Forces can be categorized into two broad groups: contact forces and non-contact forces.

Contact Forces: Forces that occur when two objects are physically touching each other.

Examples:8

- (a) Friction: The force resisting the relative motion of solid surfaces.
- (b) Tension: The force exerted by a string, rope, or wire when it is pulled tight.
- (c) Normal Force: The support force exerted upon an object in contact with a surface.

Non-Contact Forces: Forces that act at a distance without physical contact between objects.

Examples:

- (a) Gravitational Force: The attractive force between two masses, pulling them toward each other.
- (b) Magnetic Force: The force that acts between magnetic poles; can attract or repel magnetic materials.
- (c) Electric Force: The attraction or repulsion between charged particles.

Differences Between Magnetic and Gravitational Forces

S/N	FEATURES	GRAVITATIONAL FORCES	MAGNETIC FORCES
1	Nature	Always attractive	Can be attractive or repulsive
2	Source	Arises from mas	Produced by moving electric charges or magnetic materials
3	Range	Acts at a distance but with a limited range	Acts over long distances, can be felt across galaxies .
4	Strength	Generally weaker than magnetic forces.	Can be strong (e.g. refrigerator magnet).
5	Dependency	Depends on the mass of objects and distance between them	Depends on the polarity of magnets and speed of charged movement

Measurement and Calculation of Gravitational Force

The gravitational force is given by the formula mg where m equals mass in kilogram and g equals acceleration due to gravity in meters per second square

Example 1

An object of mass 5 kg falls from rest through a certain height. Determine the gravitational force. [Take $g = 10$ meters per second square]

Solution

Given :

mass of object = 5 kg.

acceleration due to gravity = 10 meters per second.

By formula,

Gravitational force = Mass x acceleration due to gravity.

$$F = m \times g.$$

$$F = 5 \text{ kg} \times 10 \text{ m/s}^2 = 50 \text{ N}.$$

Example 2.

If the gravitational force on an object is 70N and the acceleration due to gravity is 10 meters per second square, calculate the mass of the object .

Solution

Given $F = 70\text{N}$

$g = 10 \text{ m/s}^2$

mass = unknown.

By formula,

$$F = mg$$

Therefore ,

$$m = F / g$$

$$m = 70\text{N}/10 \text{ m/s}^2$$

$$m = 7 \text{ kg}$$

Example 3.

A boy lifts a 15 kg television from the floor to a table 1.2 m high. Calculate the work done by the boy.[take $g = 10 \text{ m/s}^2$]

Solution

Given:

Mass, $m = 15 \text{ kg}$

Height, $h = 1.2 \text{ m}$

By formula, Work done = mgh

$$= 15 \text{ kg} \times 10 \text{ m/s}^2 \times 1.2 \text{ m}$$

$$= 180 \text{ J}$$